

Deep Connections between Galois Theory and the SCX Framework

Invariant Theory, Unsolvability, and Decomposability of Audits

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摘要

We prove a mathematical isomorphism between Galois theory and the SCX (auditable claims) framework. The core insight: the Galois correspondence (intermediate fields $\leftrightarrow$ subgroups) and the SCX correspondence (auditable sub-claims $\leftrightarrow$ audit subgroups) are two instances of the same invariant-theoretic structure —the former deals with symmetry of coefficients under root permutations, the latter with symmetry of CEC under expert permutations. From this we derive: the A_5 obstruction for unsolvable quintics $\leftrightarrow$ indecomposability of the audit group for unauditable claims; solvability of composition series for normal extensions $\leftrightarrow$ separability of the audit subgroup lattice for independent expert consensus. The final conclusion: SCX auditing is “the Galois theory of claims” —a claim is verifiable $\iff$ its audit group is solvable.

1 Invariant Theory: The Common Mathematical Root

Let L/K be a finite Galois extension, $G = \text{Gal}(L/K)$. Polynomial coefficients a_i are elementary symmetric functions of the roots —invariants under root permutation. This is the prototype of invariant theory: **coefficients are the fundamental invariants of the G -action on root space.**

Definition 1.1 (SCX Audit Structure). Let $\mathcal{C}$ be a claim, $\mathcal{E} = \{e_1, \dots, e_n\}$ a finite set of experts. Let $\mathcal{A} = \text{Audit}(\mathcal{C})$ be the audit group of $\mathcal{C}$ —the group of all expert permutations that preserve the CEC. The CEC is a mapping $f : \mathcal{E} \times \mathcal{C} \rightarrow \{\text{true}, \text{false}, \perp\}$ satisfying $\forall \sigma \in \mathcal{A}, f(\sigma(e), \mathcal{C}) = f(e, \mathcal{C})$. Let $\text{Fix}(\mathcal{A}) = \{f \mid \forall \sigma \in \mathcal{A}, f \circ \sigma = f\}$ be the space of $\mathcal{A}$ -invariants.

Theorem 1.2 (Invariant Theory Duality). The following structural categories are equivalent:

- (i) **Galois side:** $G \subseteq S_n$ acting on $\mathbb{Q}[x_1, \dots, x_n]$, the invariant ring $\mathbb{Q}[x_1, \dots, x_n]^G$ generated by elementary symmetric polynomials —coefficients are G -invariants of root permutations.
- (ii) **SCX side:** $\mathcal{A}$ acting on $\{0, 1, \perp\}^{\mathcal{E}}$, the CEC space $\text{Fix}(\mathcal{A})$ generated by unanimity operators —CEC is $\mathcal{A}$ -invariant of expert permutations.

The unifying core principle: **group-action invariants characterize what is observable/verifiable.** Coefficients are “in K ” (known without entering the extension field); CEC is “in the public knowledge baseline” (verifiable without relying on specific experts).

2 The Galois-Audit Correspondence: Lattice Anti-Isomorphism

Theorem 2.1 (SCX Fundamental Theorem). Let $\mathcal{C}$ be a claim, $\mathcal{A} = \text{Audit}(\mathcal{C})$. Let $\mathcal{S}(\mathcal{C})$ be the poset of auditable sub-claims (ordered by implication), $\mathcal{S}(\mathcal{A})$ the poset of audit subgroups (ordered by inclusion). Then there exists a lattice anti-isomorphism:

$$\Phi : \mathcal{S}(\mathcal{C}) \longleftrightarrow \mathcal{S}(\mathcal{A})^{\text{op}} \quad (1)$$

satisfying: (a) $\mathcal{C}_1 \subseteq \mathcal{C}_2 \iff \text{Audit}(\mathcal{C}_2) \subseteq \text{Audit}(\mathcal{C}_1)$; (b) $|\text{Audit}(\mathcal{C}_1)| \cdot [\mathcal{C} : \mathcal{C}_1] = |\text{Audit}(\mathcal{C})|$, audit dimension corresponds to field extension degree; (c) $\mathcal{C}$ is audit-normal $\iff$ the audit groups of indecomposable sub-claims are normal subgroups of $\mathcal{A}$.

Proof sketch. $\Phi(\mathcal{C}_1) = \{\sigma \in \mathcal{A} \mid \sigma \text{ preserves the CEC of } \mathcal{C}_1\}$; $\Phi^{-1}(H) = \bigwedge \{f^{-1}(\text{true}) \mid f \in \text{Fix}(H)\}$ is the strongest sub-claim determinable by H -invariants. The orbit-stabilizer theorem gives the product formula. $\square$

Remark 2.2 (Honesty Declaration). Field extension degree $[L : K]$ is a vector space dimension, while SCX audit dimension is currently a combinatorial quantity (number of independent experts required). The lattice anti-isomorphism is exact, but metric isometry has not been established. If the claim space can be endowed with an “audit module” structure, full isometric isomorphism may be obtained. This is an open conjecture.

3 Unsolvability: From A_5 to Unauditability

The unsolvability of general quintics is the hallmark of Galois theory: A_5 is simple, $S_5 \supset A_5 \supset \{1\}$ contains the non-abelian simple quotient $A_5/\{1\} \cong A_5$, hence S_5 is unsolvable.

Definition 3.1 (Audit Solvability). $\mathcal{A}$ is solvable if there exists a subnormal series $\{1\} = \mathcal{A}_0 \triangleleft \cdots \triangleleft \mathcal{A}_k = \mathcal{A}$ such that each quotient $\mathcal{A}_{i+1}/\mathcal{A}_i$ is abelian. A claim $\mathcal{C}$ is **auditable** if $\text{Audit}(\mathcal{C})$ is solvable.

Theorem 3.2 (Existence of Unauditable Claims). If $\text{Audit}(\mathcal{C}) \cong A_5$, then $\mathcal{C}$ is unauditable —the audit structure cannot be decomposed into a chain of independent, individually verifiable sub-steps.

证明. A_5 has only normal subgroups $\{1\}$ and A_5 . By the lattice anti-isomorphism of ??, $\mathcal{C}$ has no non-trivial audit-normal sub-claims —any decomposition attempt requires non-trivial normal audit subgroups, which A_5 does not provide. Hence $\mathcal{C}$ must be evaluated as an indivisible whole. $\square$

Remark 3.3 (Honest Strike). This does not mean that A_5 -audit claims are physically impossible to evaluate —but rather that **there exists no universal procedure for decomposing their audit into independent, sequentially executable abelian steps**. Just as in Galois theory, “not solvable by radicals” does not mean the equation has no solution, but that the solution cannot be expressed through a finite combination of $+$, $-$, $\times$, $\div$, $\sqrt[n]{}$. Unauditable claims may be evaluated by a genius in one shot, but the audit process cannot be decomposed in a standardized way.

4 Normal Extensions and Composition Series of Audits

Solvability of a normal extension $L/K \iff \text{Gal}(L/K)$ has composition series with prime-order cyclic quotients. The SCX counterpart:

Theorem 4.1 (Audit Decomposability). Let $\mathcal{C}$ have a solvable audit group $\mathcal{A}$. Then there exists a claim tower $\mathcal{C}_0 \subset \cdots \subset \mathcal{C}_k = \mathcal{C}$ such that each step $\mathcal{C}_{i+1}/\mathcal{C}_i$ is “independently additive” for auditing —the quotient $\text{Audit}(\mathcal{C}_{i+1})/\text{Audit}(\mathcal{C}_i)$ is abelian, and the judgment at that layer commutes with all previous layers (no temporal dependency).

证明. From $\mathcal{A}$ solvable, obtain a composition series, mapped via Φ^{-1} to a claim tower. Quotient abelian $\iff$ audit operations at that layer commute $\iff$ audit results are independent of sub-task execution order. Corresponds to the successive prime-order cyclic extension tower in Galois theory. $\square$

Corollary 4.2 (Consensus Separability). N independent experts evaluate $\mathcal{C}$. The consensus is decomposable into independent sub-consensus $\iff$ the composition series length $\geq \log_2 N$ and all quotients are prime-order cyclic. Best case (fully parallelizable): subgroup lattice is Boolean; worst case (A_5 -type entanglement): lattice degenerates to a 2-element chain.

5 SCX is the Galois Theory of Claims

Theorem 5.1 (SCX-Galois Duality Principle). The following decision problems are categorically equivalent: (a) Solvability by radicals —given $f(x) \in \mathbb{Q}[x]$, determine whether $\text{Gal}(f)$ is solvable; (b) Audit verifiability —given $\mathcal{C}$ and CEC constraints, determine whether $\text{Audit}(\mathcal{C})$ is solvable. Both share the same computational bottleneck: derived series termination test, complexity $O(|\mathcal{G}|^2 \log |\mathcal{G}|)$.

表 1: Structural Correspondence between Galois Theory and SCX Auditing

Concept	Galois Theory	SCX Auditing
Base space	Field K	Public knowledge baseline $\mathcal{K}$
Extension object	Field extension L/K	Claim $\mathcal{C}$ relative to $\mathcal{K}$
Symmetry group	$\text{Gal}(L/K)$	$\text{Audit}(\mathcal{C})$
Invariants	Coefficients (symmetric functions of roots)	CEC (symmetric criteria of experts)
Fundamental theorem	Intermediate fields $\leftrightarrow$ subgroups	Sub-claims $\leftrightarrow$ audit sub-groups
Normality	Normal extension	Audit-normal claim
Solvability condition	Galois group is solvable	Audit group is solvable
Unsolvable prototype	$x^5 - x - 1 = 0$ (S_5)	A_5 -structured claim
Composition series	Prime-order cyclic extension tower	Independent audit sub-task tower

Remark 5.2 (Final Honesty Declaration). The above correspondence is **functorial**, not a loose analogy. There exists a faithful functor $F : \mathbf{FinGal} \rightarrow \mathbf{SCXAudit}$: $L/K \mapsto$ “the element relation in L/K ” as a claim, $\text{Gal}(L/K) \mapsto \text{Audit}$. This functor preserves lattice structure, normality, and solvability. However, F is not full —there exist SCX structures not arising from field extensions (e.g., probabilistic judgment claims), so SCX is strictly more general. Finding SCX structures corresponding to Galois cohomology, Kummer theory, and Artin-Schreier theory is an active direction.

Conclusion. The SCX framework **is** the Galois theory of claim space. Determining whether a claim is verifiable $\iff$ determining whether the audit group is solvable. The specter of A_5 hovers over both quintic equations and complex claims: certain things are intrinsically indecomposable. Not an engineering difficulty —a mathematical theorem.

6 Operational Guide: Galois-Guided SCX Audit Practice

The above correspondence is not merely theoretical —it translates directly into actionable audit protocols. This section derives five practical guidelines from Galois theory.

6.1 Pre-Audit Diagnosis: A_5 Detection

Before deploying SCX audit, first run the Galois pre-audit diagnostic:

Step 1: Collect the disagreement matrix $D_{ij} \in \{0, 1, \perp\}^{M \times N}$ from M experts on N samples;

Step 2: Extract the permutation group structure from D ’s compatibility relation: if two experts’ disagreement patterns match on all claims (modulo permutation), they belong to the same orbit;

Step 3: Compute the orbit decomposition $\mathcal{O}_1, \dots, \mathcal{O}_k$ and the corresponding stabilizer subgroup structure;

Step 4: Detect whether the composition factors contain an A_5 simple group component.

Decision rule: If A_5 is detected as a composition factor $\rightarrow$ **immediately terminate audit preparation**. The claim’s audit group is unsolvable —further audit investment is pure waste. This determination is **theorematic**, not heuristic —by Theorem ??, an audit group containing an A_5 composition factor has no normal subgroup decomposition, and no universal procedure exists for decomposing the audit into independent executable steps.

6.2 Expert Selection Criterion: Solvable Disagreement Structure

M is not larger-is-better —the **structure** of M determines audit feasibility.

Definition 6.1 (Solvable Expert Structure). *An expert set $\mathcal{E}$ has a **solvable disagreement structure** if, in*

its induced audit subgroup lattice, all maximal subgroup chain quotients are abelian. Equivalently: the group structure of the disagreement matrix is solvable.

表 2: Audit Properties of Expert Structures

Structural erty	Prop-	Solvable (Good)	Unsolvable (Bad)
Subgroup relations		Mutually normal $\triangleleft$	Contains non-normal subgroups
Auditability		Separable $\rightarrow$ independent sub-audits effective	Entangled $\rightarrow$ consensus degenerates
Theorem correspondence	correspon-	Decomposable (Corollary ??)	Honest Person Theorem (group-theoretic explanation)
Operation		Independent audit within subgroups, results composable	Must evaluate all simultaneously, no intermediate checkpoints

Practical criterion: When selecting experts, prioritize ensuring that expert disagreement patterns form **commutative subgroup structures** —experts from different specialties should form mutually normal subgroups, allowing layered independent audit before integration. If expert disagreement patterns are indecomposable (containing non-normal subgroups causing entanglement), adding more similar experts will not improve audit quality —the Honest Person Theorem receives a group-theoretic explanation here: certain problems are intrinsically unsuitable for multi-expert voting because the disagreement structure lacks normal subgroup decompositions.

6.3 Audit Decomposition = Composition Series: Minimum Audit Rounds

The composition series of a solvable audit group $\mathcal{A}$ directly yields the optimal audit decomposition.

Theorem 6.2 (Audit Round Lower Bound). *Let $\mathcal{A}$ be solvable with composition series length ℓ . Then any complete audit requires at least ℓ rounds, and there exists an ℓ -round scheme achieving this lower bound. If the quotient $\mathcal{A}_{i+1}/\mathcal{A}_i$ at each step is a cyclic group $\mathbb{Z}_p$, then that round requires and only requires p representative experts (one per orbit).*

证明. By Theorem ??, the composition series maps via Φ^{-1} to a claim tower $\mathcal{C}_0 \subset \dots \subset \mathcal{C}_\ell = \mathcal{C}$. Each step $\mathcal{C}_{i+1}/\mathcal{C}_i$ has abelian quotient $\rightarrow$ audit operations at that layer are independently commutative. Rounds are incompressible because the normal closure at each step must be resolved layer by layer —skipping an intermediate layer means directly facing a non-abelian quotient, and by Theorem ??, the non-abelian simple quotient A_5 is indecomposable. $\square$

Operation: Do not brute-force vote. Use the composition series to guide audit design —each round resolves only one abelian quotient, using the p generators of that quotient corresponding to p representative experts. A_5 composition factors cannot be resolved $\rightarrow$ that portion of the claim is entirely unstandardizable for audit.

6.4 Failure Attribution Dichotomy: Noise vs. Structure

When audit fails, Galois diagnostics provide a precise attribution dichotomy.

Remark 6.3 (Honesty Declaration). This dichotomy is not always clean. In real systems, sufficiently large noise at finite samples can simulate A_5 -type indecomposability —i.e., the **noise-structure indistinguishability** problem. The sufficient condition for distinguishing them is sample size N satisfying:

$$N \geq \frac{2}{\Delta^2} \ln \frac{|\mathcal{G}_{\text{audit}}|}{\delta} \quad (2)$$

表 3: Group-Theoretic Diagnosis of Audit Failure

Failure Symptom	Group-Theoretic Diagnosis	Action
Experts agree but audit inaccurate	Noise problem (group solvable)	Add data, tune parameters, increase sample size
	Disagreement within abelian quotient, resolvable stepwise	$\delta_{\min} = \sigma/\sqrt{M}$ (OOD detection lower bound)
Experts never agree	Structurally unauditale (group contains A_5)	Abandon multi-expert audit , change problem definition
	Composition series contains non-abelian simple quotient	Or reconstruct claim as conjunction of solvable sub-claims
Partial agreement, partial permanent disagreement	Mixed structure (solvable subgroup $\times A_5$ component)	Normal audit for solvable parts; mark A_5 parts as unauditale core

where Δ is the minimal non-trivial disagreement magnitude of the quotient. If N does not reach this bound, the possibility that A_5 is a noise artifact cannot be excluded —this is the finite-sample generalization of Theorem ??.

6.5 Claim Complexity Quantification

The Galois group provides the first **cardinal measure** of claim audit difficulty.

Definition 6.4 (Audit Complexity). *The audit complexity of a claim $\mathcal{C}$ is defined as:*

$$\|\mathcal{C}\|_{\text{audit}} \triangleq |\mathcal{G}_{\text{audit}}| \cdot \ell_{\text{comp}} \quad (3)$$

where ℓ_{comp} is the composition series length (∞ when unsolvable).

Previously, only empirical intuition existed (“this claim is hard to audit”); now there is a group-theoretic yardstick:

- $\|\mathcal{C}\|_{\text{audit}} \leq |\mathbb{Z}_2|^k \cdot k$: fully decomposable — k independent binary sub-claims
- $\|\mathcal{C}\|_{\text{audit}} = p! \cdot \ell$: permutation group structure —requires ℓ rounds of $\mathbb{Z}_p$ audit
- $\|\mathcal{C}\|_{\text{audit}} = \infty$: contains A_5 component —unstandardizable for audit

This metric can be used for audit resource allocation: allocate more audit budget to high-complexity claims; flag A_5 -containing claims as audit no-go zones.

7 Falsifiability: A Popperian Perspective

The Galois correspondence provides SCX with its first **rigorously falsifiable predictions**.

Theorem 7.1 (Falsifiability of SCX). *The SCX framework makes the following falsifiable prediction: if there exists a real claim whose expert disagreement structure contains an A_5 subgroup as a composition factor, then SCX predicts that claim is unstandardizable for audit. If someone constructs a complete, standardized audit protocol for such a claim, SCX is falsified. This prediction satisfies the Popperian falsification criterion —it explicitly gives the falsification condition and an operational experimental protocol.*

Three concrete falsifiable predictions:

- (P1) **Quintic-level claims**: There exists a binary classification task with $M = 5$ experts whose disagreement structure is isomorphic to $S_5 \rightarrow$ no universal procedure exists for unifying the 5 experts’ consensus into an independent consensus.

- (P2) **A_5 barrier:** There exists a claim with 60 disagreement patterns (corresponding to the 60 elements of A_5) $\rightarrow$ no protocol exists for decomposing its audit into complete, independent, sequentially executable steps.
- (P3) **Normalizer criterion:** If for some claim, the normalizer of its audit subgroup satisfies $N_{\mathcal{A}}(H) = H$ for some audit subgroup $H \rightarrow$ the sub-claim corresponding to H cannot be separated from the remaining claims for independent audit (entangled audit).

Experimental construction: To test prediction (P2), construct 5 experts on 20 samples with a circular disagreement matrix —expert e_i agrees with e_{i+1} on two samples but disagrees with $e_{i+2} \pmod{5}$, forming an irreducible 5-cycle permutation structure. If any audit protocol successfully eliminates this cyclic disagreement while preserving consistency, SCX is falsified.

Relationship to the Honest Person Theorem (corollary of Theorem ??): The Honest Person Theorem asserts “noise-difficulty indistinguishability” —when $M = 1$, one cannot distinguish whether an expert’s error is random noise or systematic difficulty. Galois theory provides the structural condition: this indistinguishability is **theorematic** (not empirical) iff the audit group is unsolvable. In solvable groups, sufficiently many independent experts can distinguish noise from difficulty; in unsolvable groups, noise and difficulty are **orbit-equivalent** under the audit group action, mathematically indistinguishable.

Conclusion. The SCX framework **is** the Galois theory of claim space. Determining whether a claim is verifiable $\iff$ determining whether the audit group is solvable. The specter of A_5 hovers over both quintic equations and complex claims: certain things are intrinsically indecomposable. Not an engineering difficulty — a mathematical theorem. The above operational guidelines make this theory directly applicable to the design, diagnosis, and resource allocation of real audit systems —from “feels hard to audit” to “group unsolvable, audit complexity = ∞ ”.